

Methodology for Cloud Fraction Analysis

1. Data Collection

The study uses satellite observations and CMIP6 climate model simulations to analyze cloud fraction variability over the period **1990–2025**.

Observational datasets

ISCCP HGM

- Time period: **1990–2017**
- Variable used: **Cloud Amount (`cldamt`)**
- Temporal resolution: Monthly
- Purpose: Long-term historical cloud fraction observations

CERES SYN1deg Edition 4.2

- Time period: **2001–2025**
- Variable used: **Total Cloud Area Fraction (`cldarea_total_mon`)**
- Temporal resolution: Monthly
- Purpose: Modern satellite observations with improved calibration
-

Climate model datasets (CMIP6)

Three CMIP6 models were analyzed:

- UKESM1-0-LL
- IPSL-CM6A-LR
- MRI-ESM2-0

Historical simulations and SSP245 simulations were combined to obtain continuous model records extending to 2025.

2. Data Pre-processing

Before analysis, all datasets were standardized.

The following preprocessing steps were performed:

- NetCDF files were imported using **xarray**
- Monthly cloud fraction variables were extracted
- Time coordinates were converted into **DatetimeIndex**
- Duplicate timestamps were removed
- Missing values were handled
- All datasets were restricted to the required study period

Global Mean Cloud Fraction Calculation

Monthly global cloud fraction was calculated from each dataset.

For CMIP6 models:

- Cloud fraction variable: clt

For ISCCP:

- Cloud fraction variable: cldamt

For CERES:

- Cloud fraction variable: cldarea_total_mon

The global mean was obtained by averaging over the spatial dimensions.

Harmonization of ISCCP and CERES

Since the two satellite datasets overlap during **2001–2017**, this period was used for validation and harmonization. The following statistical comparison was performed.

Extract overlap period

2001–2017

from both datasets.

Mean Bias Correction

Instead of applying regression, a **mean bias correction** was used because it preserves the temporal variability of ISCCP while aligning it with CERES.

The corrected ISCCP values were calculated as

Corrected ISCCP

Original ISCCP

Mean Bias

CERES – ISCCP (average over 2001–2017)

Create Harmonized Dataset

The final continuous observational dataset was constructed as:

1990–2017

↓

Bias-corrected ISCCP

2018–2025

↓

CERES

The two datasets were concatenated to produce a single continuous observational record covering 1990–2025.

Global Mean Cloud Fraction

Methodology

Cloud fraction was obtained from the ISCCP, CERES, and CMIP6 datasets. The global mean cloud fraction for each month was calculated by averaging cloud fraction over all grid cells (or equal-area cells for ISCCP).

Mathematical Expression

For equally weighted grids (ISCCP HGM equal-area cells):

$$CF_{global}(t) = \frac{1}{N} \sum_{i=1}^N CF_i(t)$$

- $CF_i(t)$ = cloud fraction at grid cell i
- N = total number of grid cells
- t = monthly time

For latitude–longitude grids (CERES and CMIP6), area weighting is recommended:

$$CF_{global}(t) = \frac{\sum_{i=1}^N CF_i(t) \cos(\phi_i)}{\sum_{i=1}^N \cos(\phi_i)}$$

where

- ϕ_i = latitude of grid cell i

Twelve-Month Running Mean

To reduce short-term fluctuations and highlight long-term variability, a 12-month running mean was applied.

Rolling window:12 months

Centered averaging:True

This smoothing was applied to

- Harmonized observations
- UKESM
- IPSL
- MRI

To remove short-term variability, a centered 12-month running mean was applied.

Mathematical Expression

$$CF_{12}(t) = \frac{1}{12} \sum_{k=-5}^6 CF(t+k)$$

where

- $CF(t)$ = monthly cloud fraction

7. Monthly Climatological Seasonal Cycle

The climatological seasonal cycle was calculated by grouping all monthly observations according to calendar month.

January values

↓

Average

February values

↓

Average

December values

↓

Average

This produced the long-term monthly climatology

Mathematical Expression

$$\overline{CF}_m = \frac{1}{n_m} \sum_{y=1}^{n_m} CF(m, y)$$

where

- m = calendar month
- n_m = number of years

Cloud Fraction Anomalies

Monthly anomalies were calculated relative to the climatological seasonal cycle.

Anomaly The anomaly was computed by subtracting the climatological monthly mean from the observed monthly cloud fraction.

Monthly Cloud Fraction – Monthly Climatology

This removes the seasonal cycle and highlights interannual variability.

Mathematical Expression

$$CF'(t) = CF(t) - \overline{CF}_m$$

where

- $CF(t)$ = observed cloud fraction
- $\overline{CF}_m$ = climatological monthly mean

Tropical Mean Cloud Fraction

The tropical analysis was performed using only grid cells located between 30°S

and

30°N.

The same procedures were repeated:

- Global mean calculation
- Running mean
- Monthly climatology
- Anomalies

Mathematical Expression

$$CF_{trop}(t) = \frac{\sum CF_i(t) \cos(\phi_i)}{\sum \cos(\phi_i)}$$

where

$$-30^\circ \leq \phi \leq 30^\circ$$

Model Bias Analysis

Bias for each CMIP6 model was calculated by subtracting the harmonized observational dataset from the corresponding model values.

Bias

Model

Observation

Positive bias

↓

Model overestimates cloud fraction

Negative bias

↓

Model underestimates cloud fraction

Bias quantifies the difference between model simulations and observations.

Mathematical Expression

$$Bias = CF_{model} - CF_{obs}$$

Interpretation:

- Bias > 0 → model overestimates cloud fraction.
- Bias < 0 → model underestimates cloud fraction.

Harmonization (Mean Bias Correction)

Methodology

A mean bias correction was applied to align ISCCP with CERES while preserving ISCCP's temporal variability.

Mathematical Expression

$$CF_{harmonized} = CF_{ISCCP} + Bias_{mean}$$

where

$$Bias_{mean} = \frac{1}{N} \sum (CERES - ISCCP)$$

For your analysis:

$$Bias_{mean} = 2.483\%$$

Thus,

$$CF_{harmonized} = CF_{ISCCP} + 2.483$$

Final Harmonized Observational Record

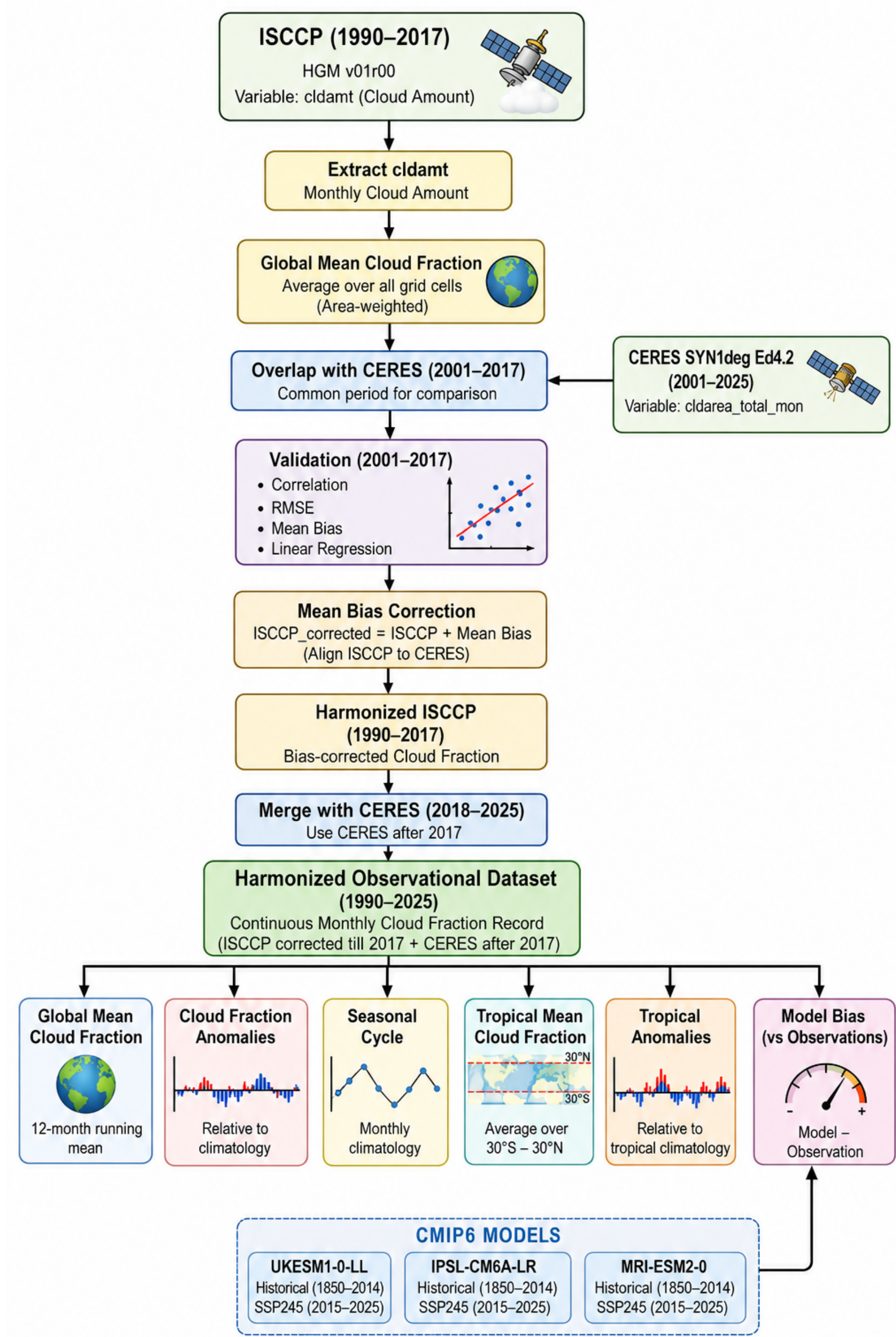
Methodology

The final continuous observational dataset was constructed by combining ISCCP and CERES.

Mathematical Expression

$$CF_{final}(t) = \begin{cases} CF_{ISCCP} + Bias_{mean}, & 1990 \leq t \leq 2017 \\ CF_{CERES}, & 2018 \leq t \leq 2025 \end{cases}$$

Workflow flowchart



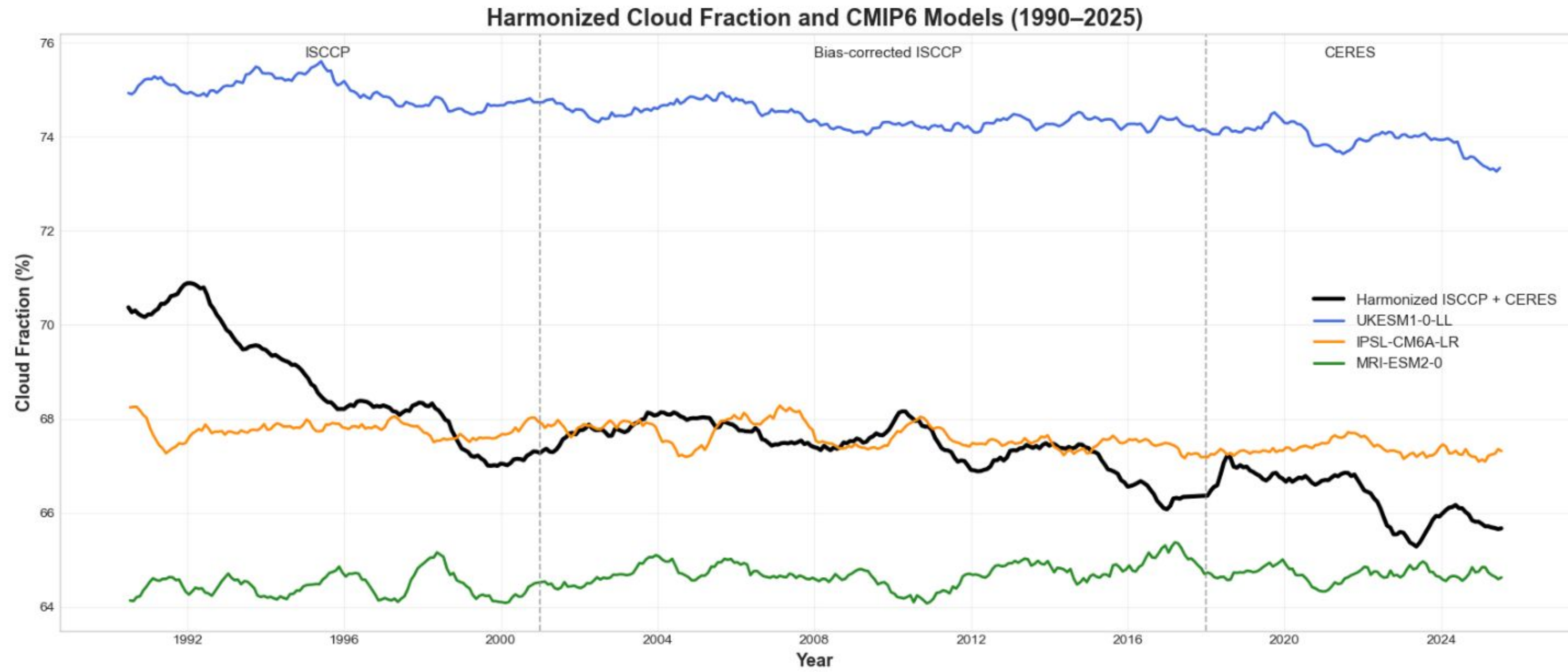
Visualization

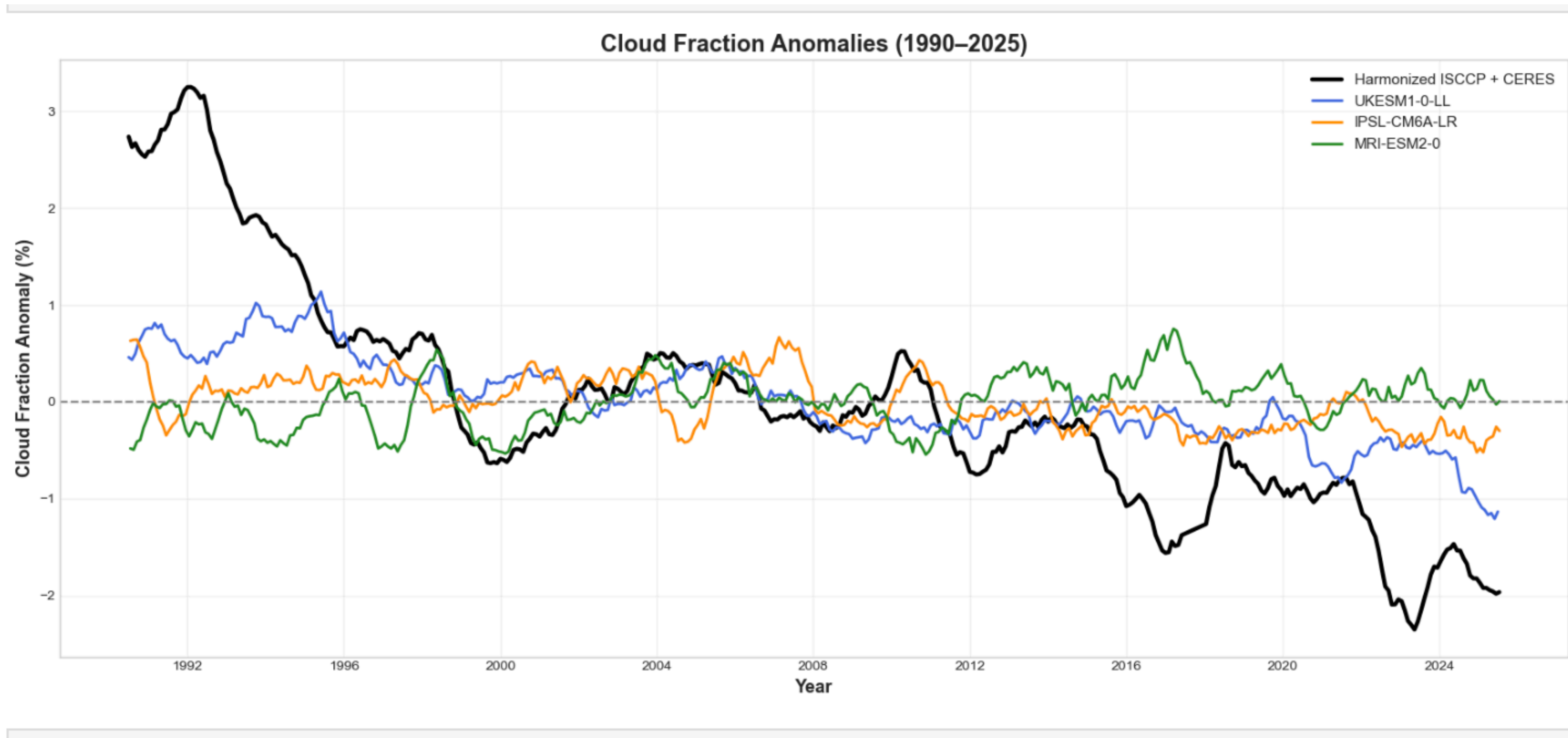
All figures were created using Matplotlib.

The figures include:

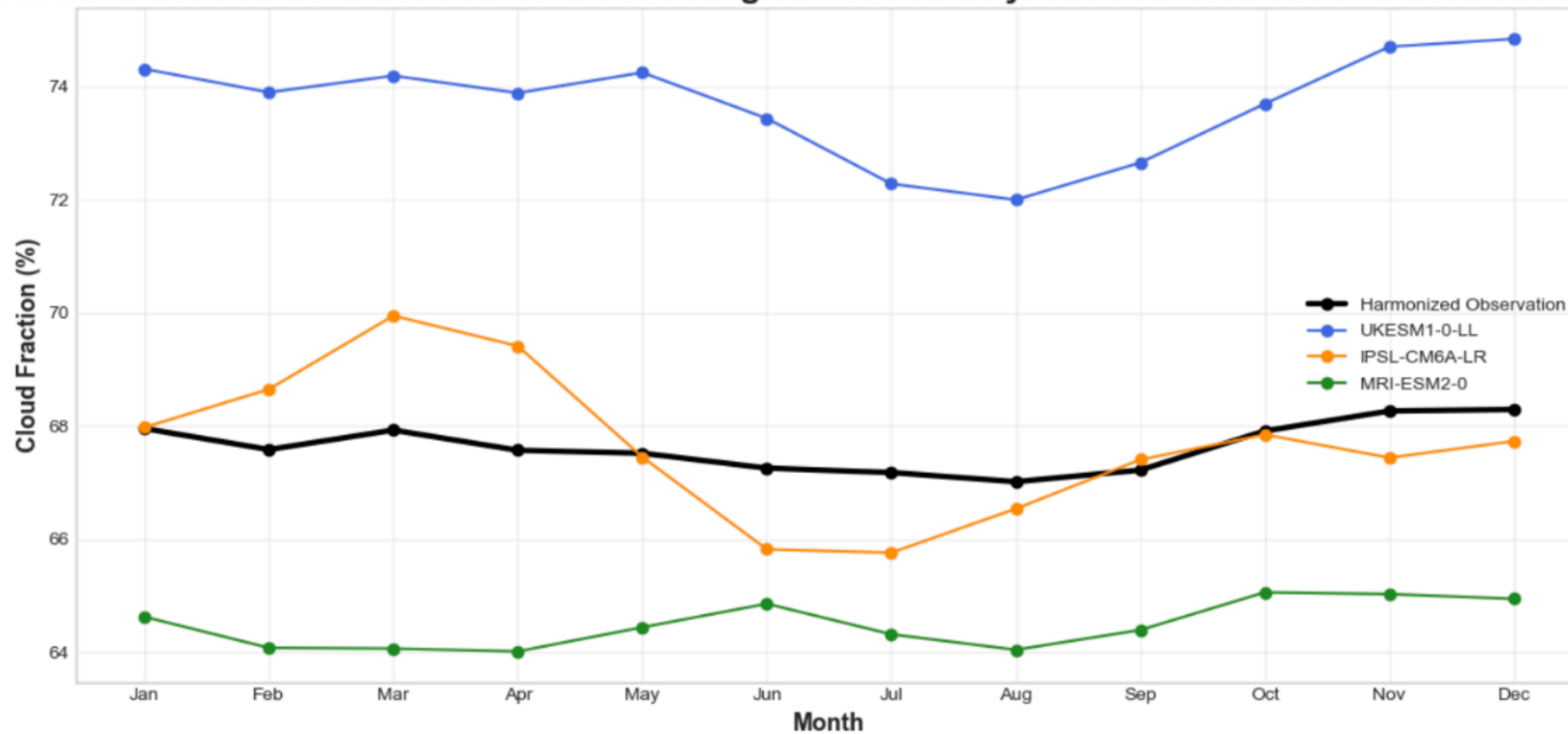
- Harmonized Cloud Fraction (1990–2025)
- Global Mean Cloud Fraction
- Cloud Fraction Anomalies
- Monthly Climatological Seasonal Cycle
- Tropical Mean Cloud Fraction
- Tropical Cloud Fraction Anomalies
- Tropical Seasonal Cycle
- CMIP6 Model Bias

● Harmonized Cloud Fraction (1990–2025)

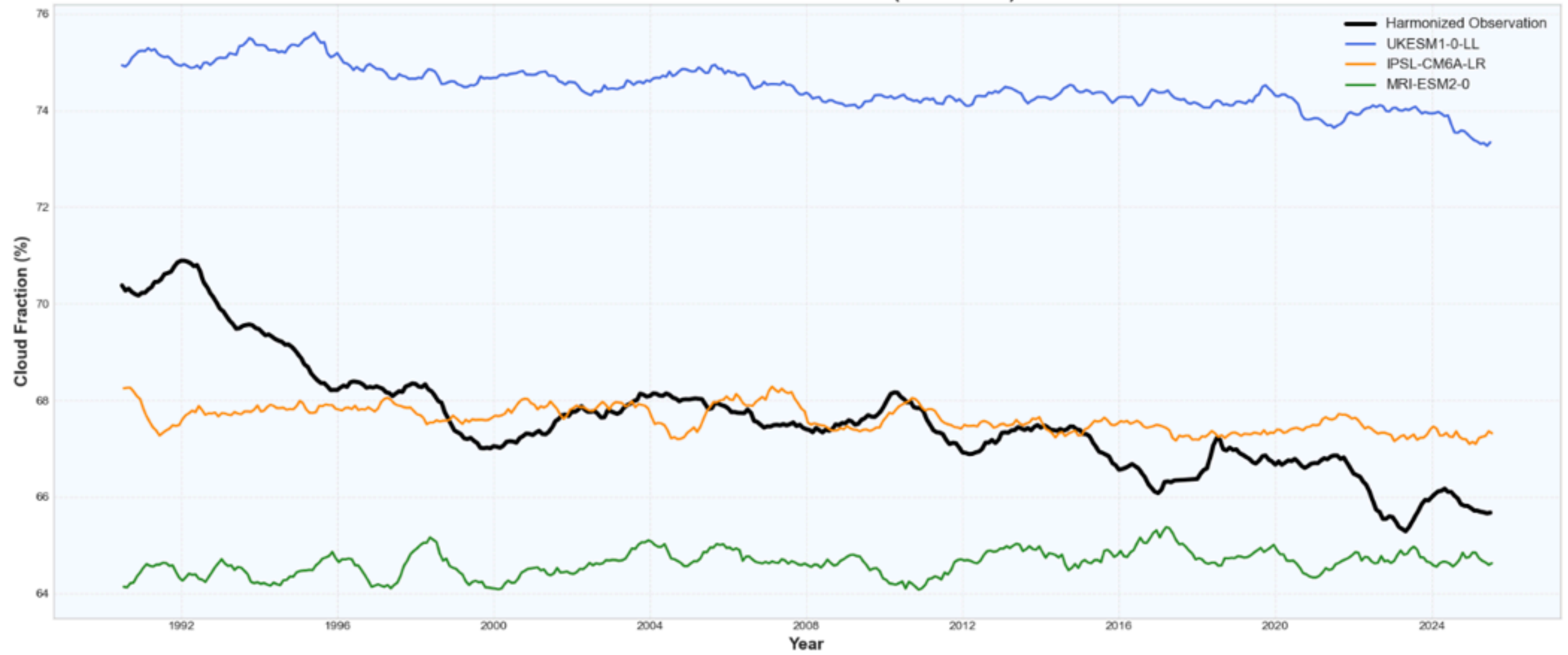




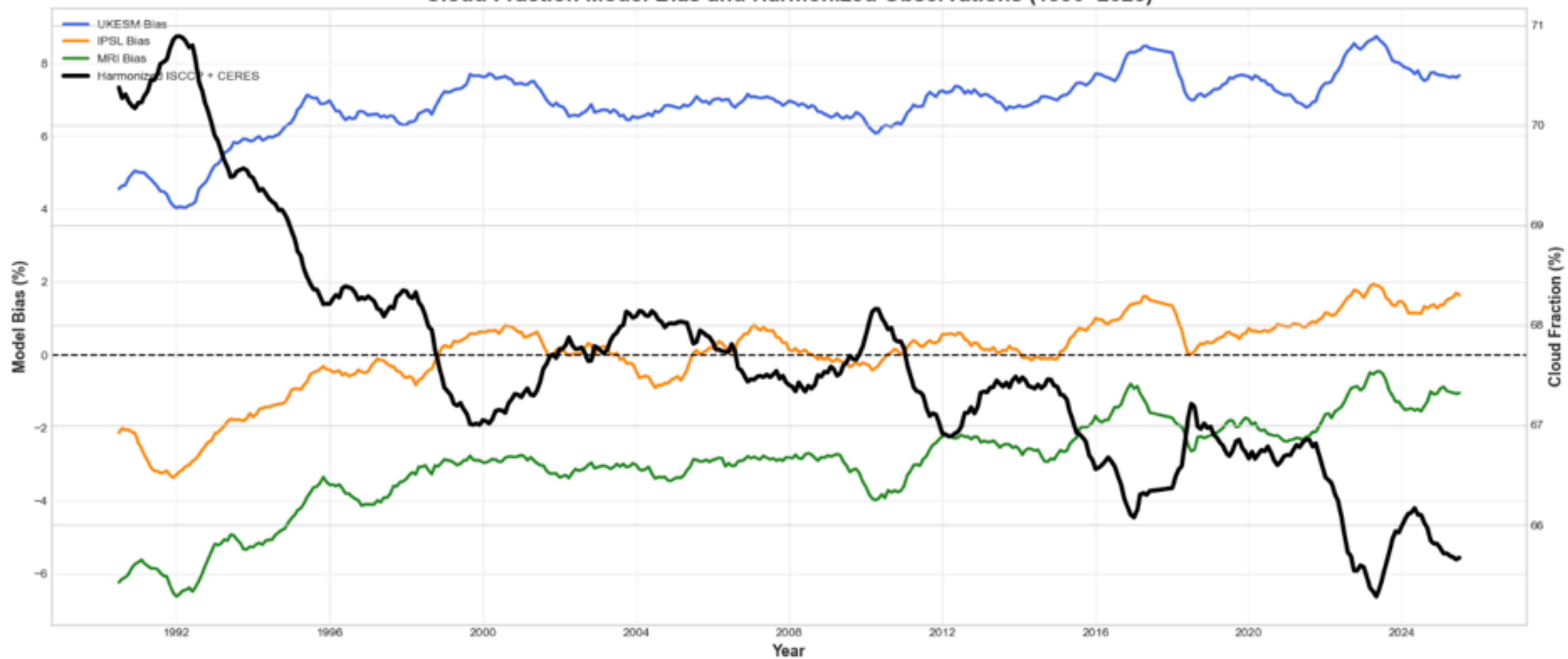
Climatological Seasonal Cycle



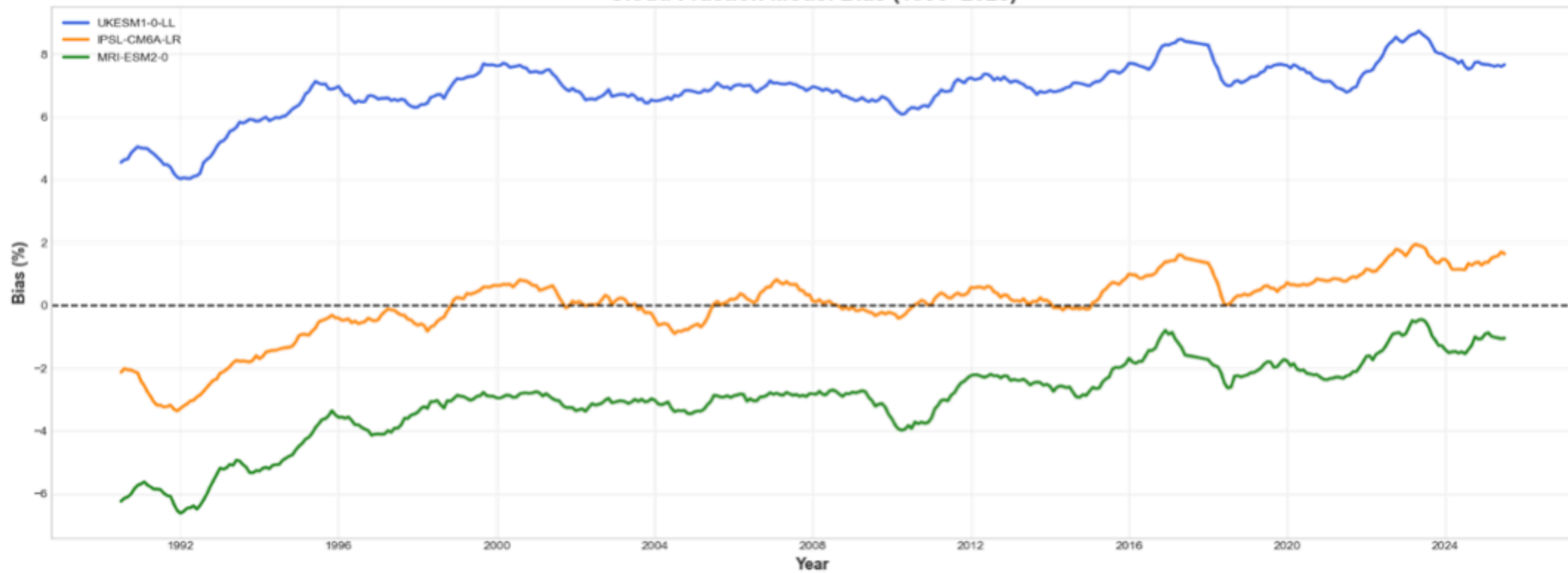
Global Mean Cloud Fraction (1990–2025)



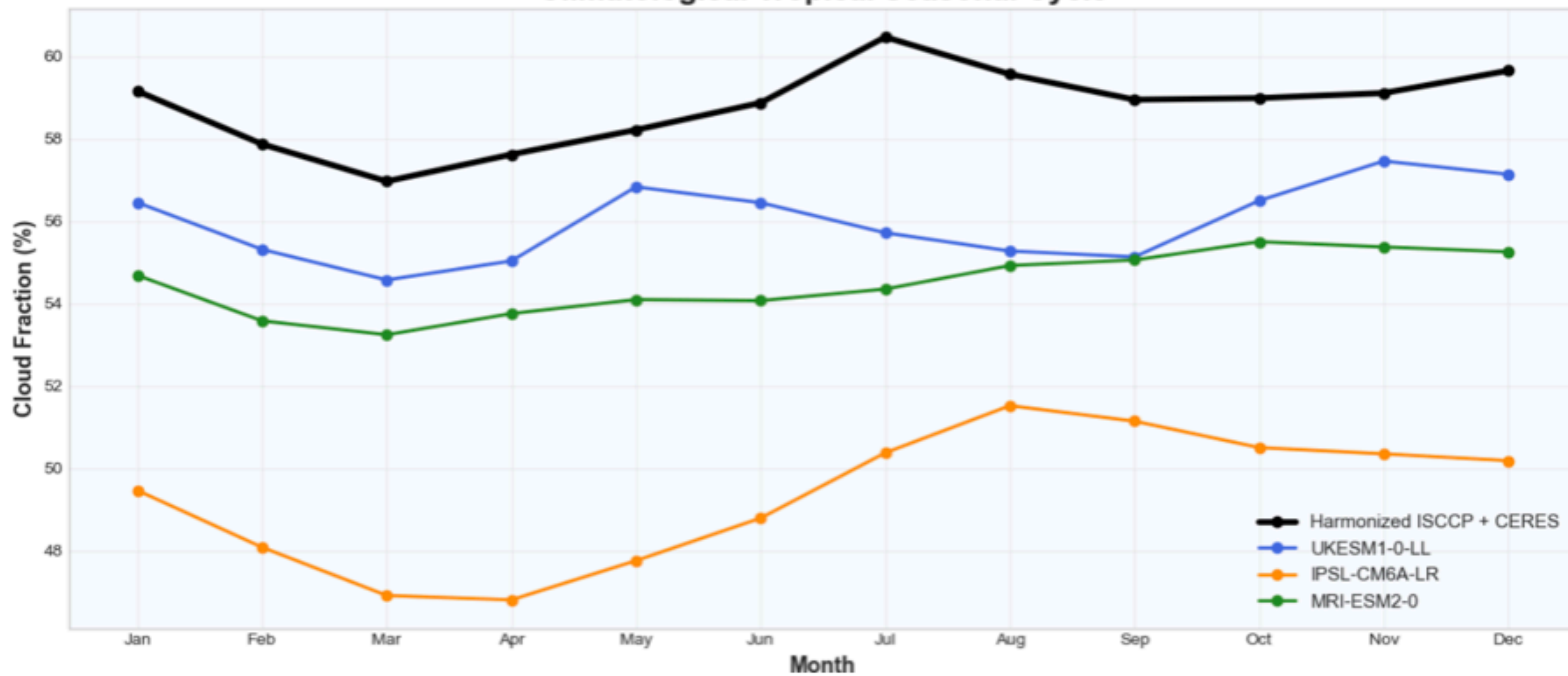
Cloud Fraction Model Bias and Harmonized Observations (1990–2025)



Cloud Fraction Model Bias (1990–2025)



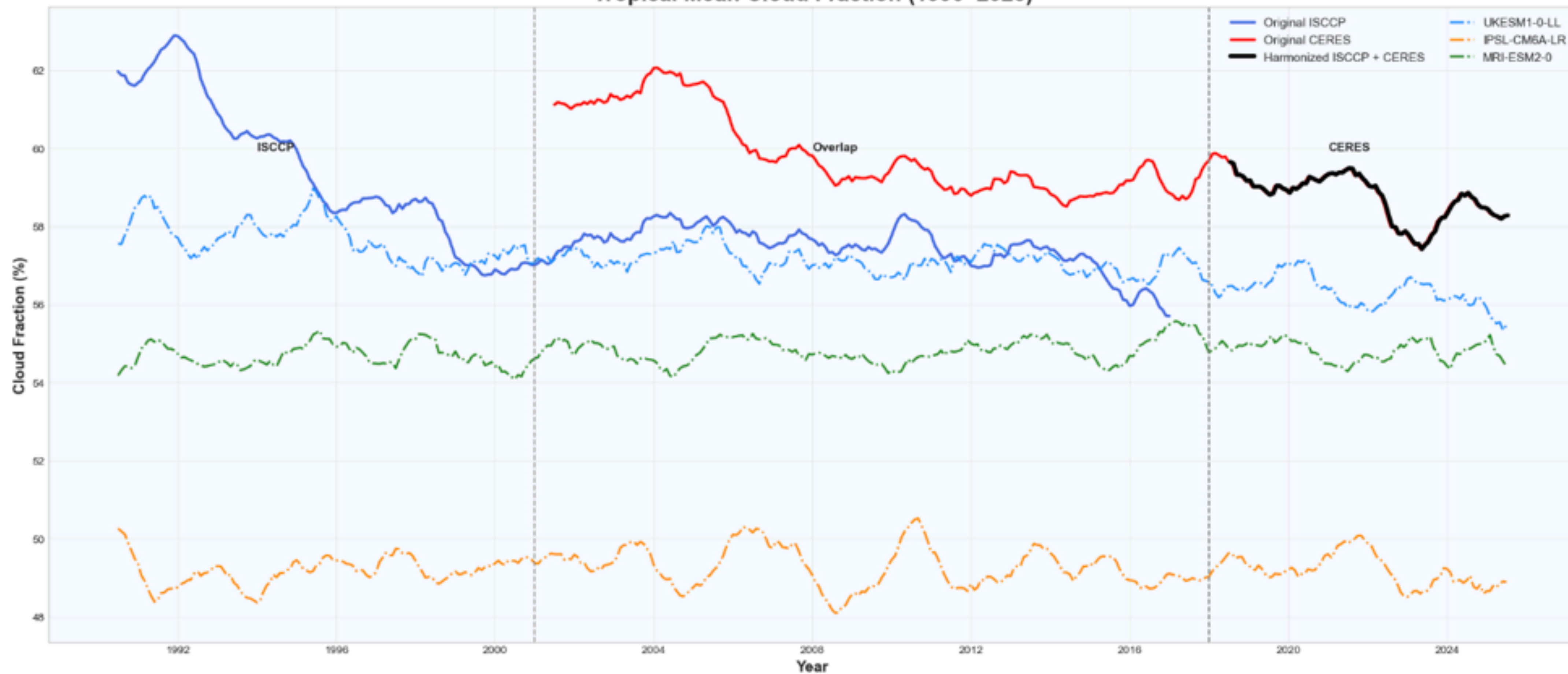
Climatological Tropical Seasonal Cycle



Tropical Cloud Fraction Anomalies (1990–2025)

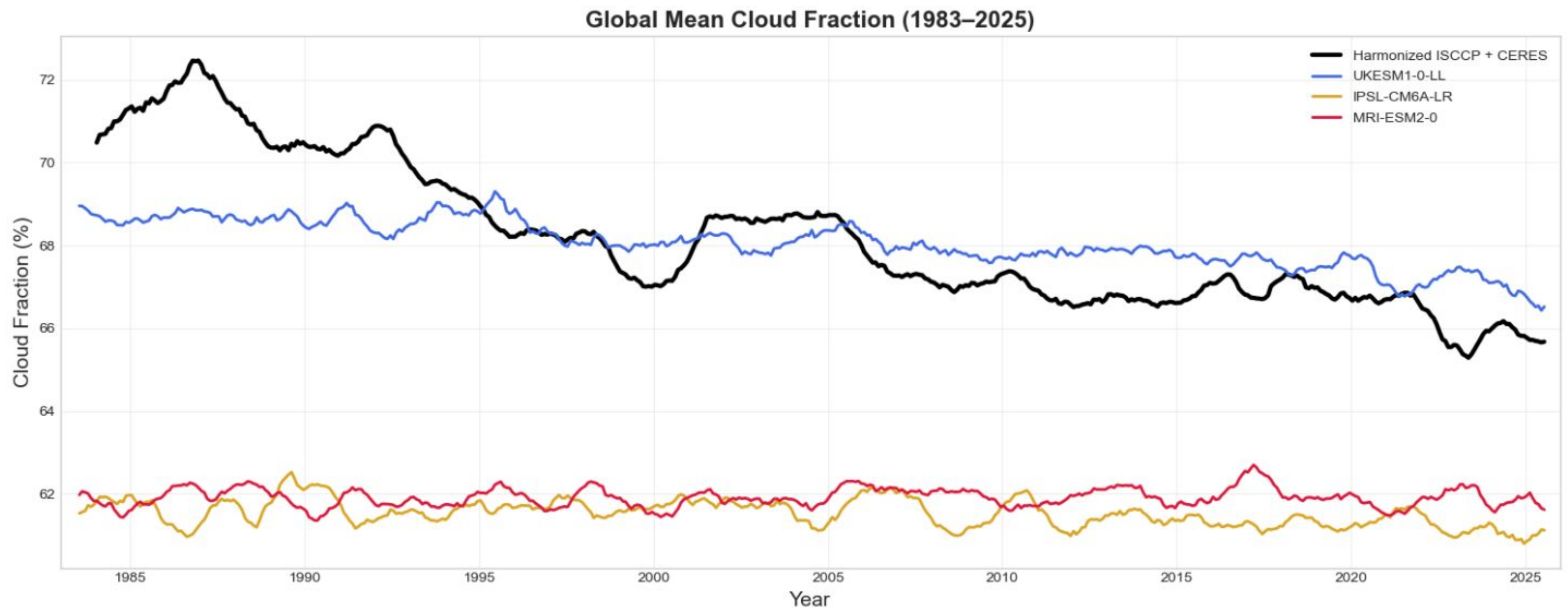


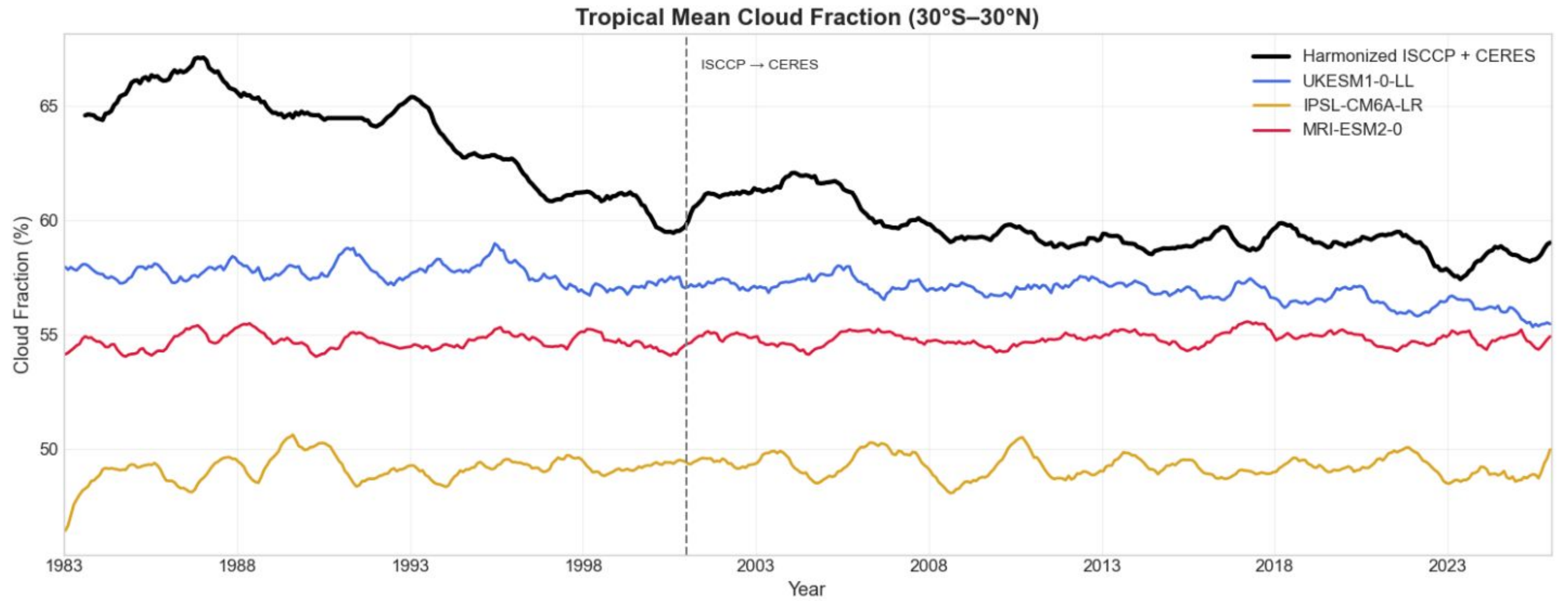
Tropical Mean Cloud Fraction (1990–2025)



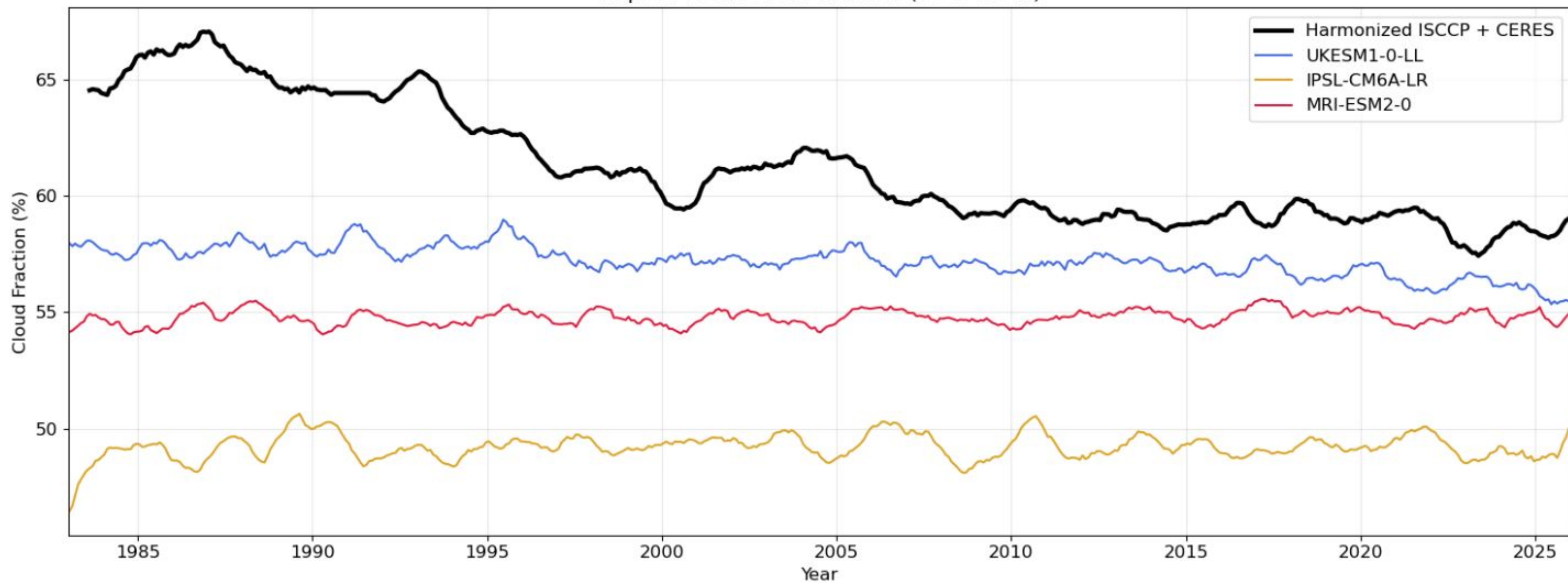
Harmonized Cloud Fraction Time Series (1990–2025)



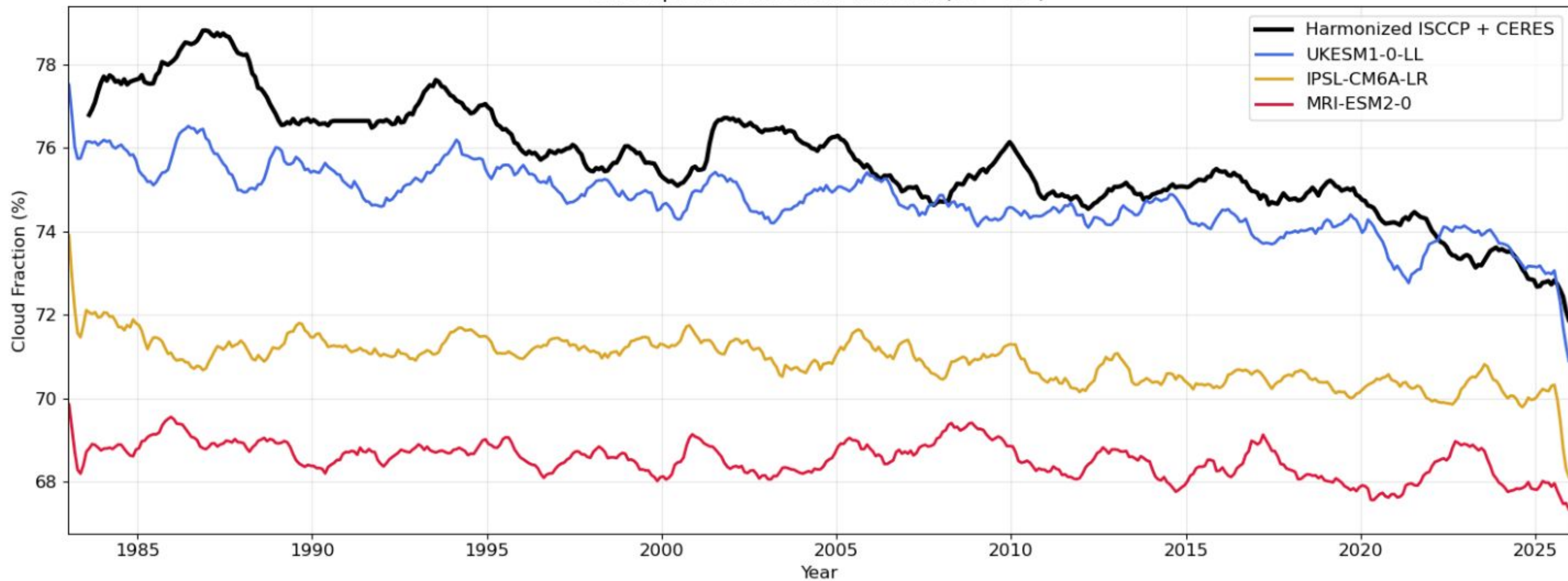




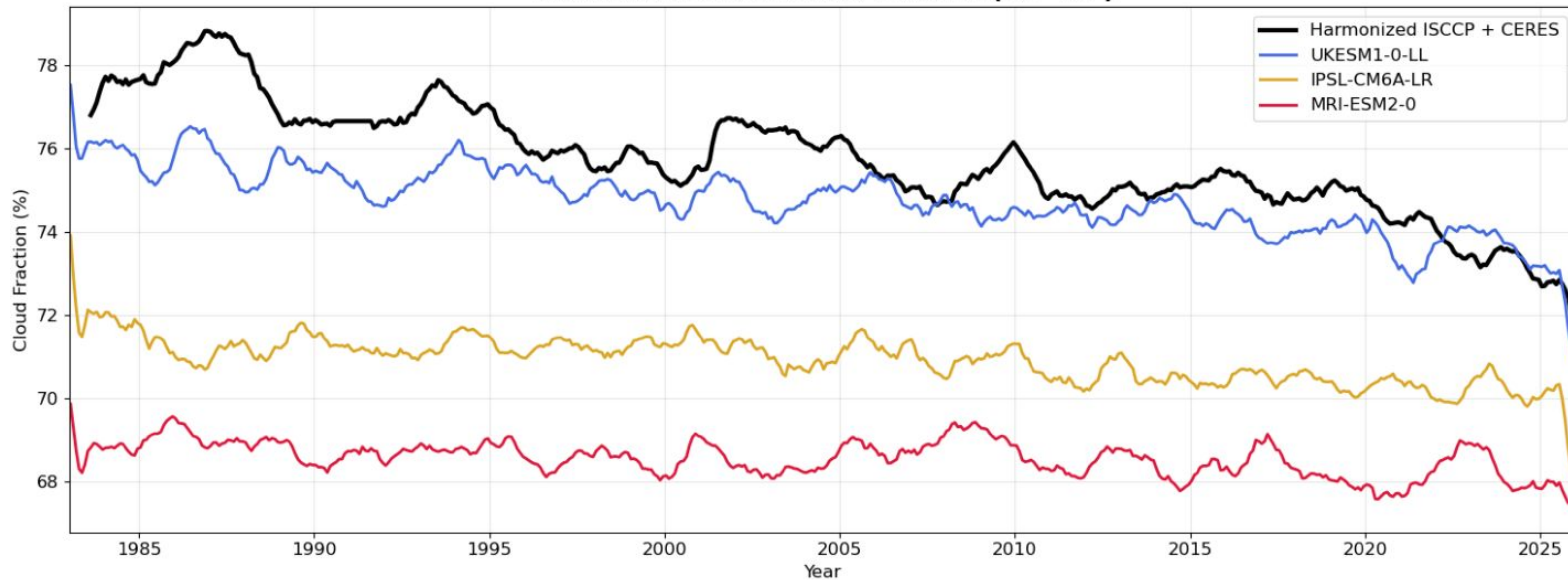
Tropical Mean Cloud Fraction (30°S-30°N)



Subtropical Mean Cloud Fraction (30°–60°)



Mid-Latitude Mean Cloud Fraction (30°-60°)



Polar Mean Cloud Fraction (60°-90°)

